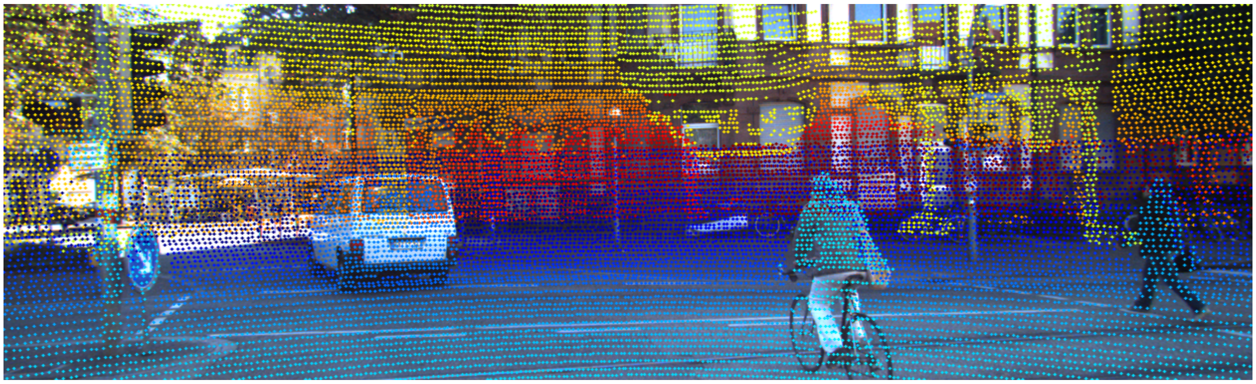
CS 588 MP0: Data Loading, Visualization & Calibration

Task 1: Questions: Observe the two images carefully. In the cam3 overlay image, why does it appear that there are missing points, especially to the side of the cars and cyclists? Why that does not happen in cam2 overlay image?

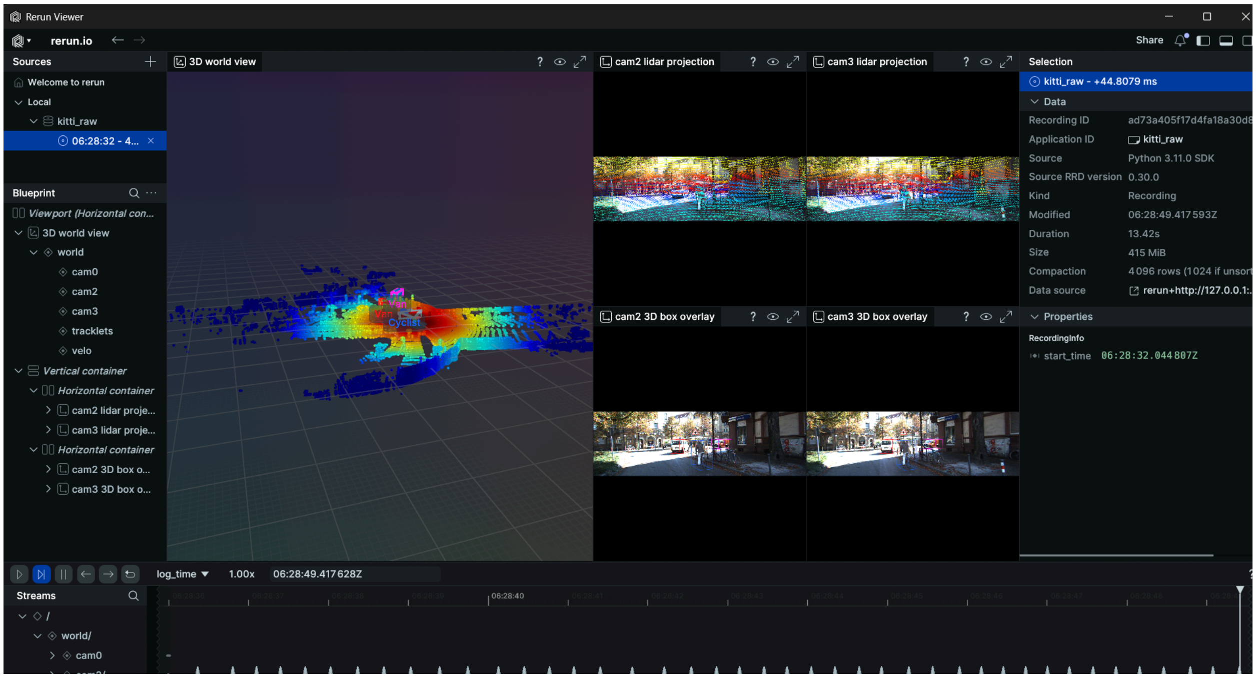
Cam2 overlay



Cam3 overlay

Ans: In the cam3 overlay, the missing points are mainly because of the physical offset between the Lidar sensor and the camera. Since the Lidar is mounted on the top center of the roof, its line of sight is slightly different from the cameras. Cam2 the left camera is positioned much closer to Lidar's center, so their views overlap pretty well. However, because Cam3 the right camera is shifted further away, it ends up seeing the blind spots that the Lidar can't reach. This happens because the Lidar pulses hit objects like cars or cyclists at an angle and the camera can see the whole side of the object, but the Lidar can't see around the corner or behind the object's body. The reason this isn't an issue with cam2 overlay is that Cam2 is the primary camera. It is specifically aligned to have the best possible overlap with the Lidar's field of view.

Task 2: Screenshot of your rerun visualization panels layout



Task 3.1: Question: When looking at the depth visualization in 3D, we can observe a lot of trailing points around the vehicle/pedestrian, making them look stretched out. Why does that happen?

Ans: The trailing points around the van and cyclists are mostly because of edge bleeding. In the disparity map, the edges of these objects aren't perfectly sharp. The pixels bleed into the background. In the 3D view, the algorithm interprets these blurry edges as points that exist halfway between the foreground and the background.

This creates a curtain of fake points that physically connects a close object to a far one in 3D space. Since the two cameras are in different spots, one can see a bit of the area behind the object while the other cannot. This makes it difficult for the computer to calculate an exact depth, so it ends up averaging the two values. This curtain of averaged points is what makes the objects look like they are trailing or stretched out into the distance.

Task 3.2 Ans:
Initial (INIT) rotation error and translation error vs ground truth:

- Rotation Error: 0.851 deg
- Translation Error: 1.314 m

ICP rotation error and translation error vs ground truth:

- Rotation Error: 0.603 deg
- Translation Error: 0.161 m

Combined pose error (rotation + translation) for both INIT and ICP:

- **INIT:** 2.165 (0.851 + 1.314)
- **ICP:** 0.764 (0.603 + 0.161)

ICP inlier ratio and outlier ratio:

- Inlier Ratio: 0.7258
- Outlier Ratio: 0.2742